

Ethan A. Wideman

Asheville, NC | 22eawideman@gmail.com | (765) 274-9965 | linkedin.com/in/ethan-wideman-412446271

EDUCATION

Indiana University – Luddy School of Informatics — Bloomington, IN

May 2026

B.S. Computer Science (Software Development)

TECHNICAL SKILLS

Languages: Java, JavaScript, TypeScript, Python, C, HTML/CSS

Frameworks & Libraries: React, Next.js, Node.js, Express.js, Spring Boot, NumPy, Pandas, Matplotlib

Databases & Data: SQL, PostgreSQL, Redis

Cloud & Tools: AWS (EC2, S3, CloudFront, RDS, Batch, Route53), Docker, Kubernetes, Git, FFmpeg, HLS, TUS

PROJECTS

Neural Collaborative Filtering Recommendation System (Python, Numpy, Pandas, Matplotlib)

Summer 2026

- Implemented a **Neural Collaborative Filtering** (NCF) recommendation system from scratch in **NumPy**, including matrix factorization, MLP forward/backward passes, and the Adam optimizer — no ML frameworks used
- Trained on MovieLens 1M (1 million ratings, 6,000 users) with **temporal leave-one-out splitting** and **negative sampling**, achieving HR@10 of 0.667 and NDCG@10 of 0.402, comparable to published NCF benchmarks
- Engineered a sparse **Adam optimizer** for embedding updates using scatter-add gradient accumulation, improving HR@10 by 47% over a popularity baseline and reducing training loss from 0.693 to 0.196 over 40 epochs
- Developed an interactive demo using **Matplotlib** to visualize how recommendations shift as a user's watch history grows, comparing model output against a popularity baseline across a 100-candidate evaluation set

Streamlair — Full-Stack Video Streaming Platform (Next.js, Node.js, PostgreSQL, AWS, Kubernetes)

Spring 2026

streamlair.net

- Developed and deployed a full-stack video streaming platform using **Next.js (TypeScript)**, **Node.js/Express**, and **PostgreSQL on AWS**, supporting end-to-end video upload, processing, and playback
- Designed and implemented a **scalable video processing pipeline** using AWS Batch + FFmpeg on EC2 Spot instances to transcode videos into **HLS (360p–1080p)**, **offloading compute-intensive processing from the API server and preventing request stalls**
- Implemented **resumable uploads (TUS)** supporting video files up to **4 GB**, with real-time progress tracking and integrated **HLS playback (hls.js)** with automatic quality switching
- Designed responsive frontend with **SSR + client interactivity**, including infinite scroll feeds, full-text search, and dynamic video pages
- Architected backend with **Redis + BullMQ** job queues, **Google OAuth authentication**, and transactional email workflows
- Integrated **S3 + CloudFront CDN** for scalable low-latency media delivery
- Containerized services with **Docker** and validated scalability using **Kubernetes (HPA)** and **k6 load testing**, handling 500 concurrent virtual users

Studymate — Full-Stack LMS Web Application (React, Next.js, Express, PostgreSQL)

Spring 2025

- Collaborated on a team to build a **full-stack learning management system (LMS)** supporting course creation, messaging, and grading workflows
- Developed **RESTful APIs** using **Node.js/Express** and implemented database operations with **PostgreSQL**
- Built responsive frontend components with **React (Next.js)** and **Tailwind CSS** for an intuitive user experience
- Implemented **CI/CD** pipelines with **GitHub Actions** to automate testing and deployment workflows

WORK EXPERIENCE

Wendy's of Bloomington, Bloomington, IN

Nov 2024 - Nov 2025

Crew Member

- Delivered fast, accurate service in a **high-volume environment serving hundreds of customers daily**
- Maintained **quality and speed** while preparing concurrent orders under time constraints
- Collaborated across roles to **support team efficiency** during peak hours

McDonald's, Elwood, IN

Oct 2020 - July 2022

Crew Member / Crew Trainer

- Trained and onboarded** new employees on POS systems, food safety, and operations
- Coordinated team workflow** during peak hours to maintain service **speed and accuracy**